

Parul Institute of Engineering & Technology

Electronics & Communication Engineering Department

ASSIGNMENT-1

Subject Name: DIGITAL ELECTRONICS

Subject Code: 303105220

1. Convert the Decimal Number 250.5, 87.876 to base 3, base 4, base 7 & base 16.
2. Convert the decimal number 225.225, 865.987 to binary, octal and hexadecimal.
3. Represent the decimal number 8620 in BCD , Excess-3 , and Gray code
4. Convert the following Numbers as directed:
 - (a) $(52)_{10} = ()_2$
 - (b) $(101001011)_2 = ()_{10}$
 - (c) $(11101110)_2 = ()_8$
 - (d) $(68)_{10} = ()_{16}$
5. Define: Digital System.
6. Convert following Hexadecimal Number to Decimal :
B28, FFF, F28
7. Convert following Octal Number to Hexadecimal and Binary:
414, 574, 725.25
8. Convert the following numbers to decimal
 - (i) $(10001.101)_2$ (ii) $(101011.11101)_2$ (iii) $(0.365)_8$
 - (iv) A3E5 (v) CDA4 (vi) $(11101.001)_2$ (vii) B2D4
9. Perform the operation of subtractions with the following binary numbers using 2' s complement
 - (i) $10010 - 10011$ (ii) $100 - 110000$ (iii) $11010 - 10000$
10. Perform the operation of subtractions with the following binary numbers using 1' s complement
 - (i) $100110 - 10011$ (ii) $1000 - 110000$ (iii) $110101 - 10000$
11. Give full form for following abbreviations and explain:
 - (i) ASCII
 - (ii) EBCDIC
12. Explain weighted binary codes with examples.
13. Find 1's and 2's complement of following binary nos.
 $(10001.101)_2$ (ii) $(101011.11101)_2$
14. Find 9's and 10's complement of following binary nos.
3405.65, 87.76